Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/04/2007
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Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Due to NASA cuts and the decision to remove the research from Flight Status, funding for the co-PIs Robert Hyers and Anup Gangopadhyay was cut and they will be removed as PIs.		
COI Name (Institution):	Herlach, Dieter Ph.D. (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation) Gangopadhyay, Anup (Washington University) Hyers, Robert Ph.D. (University of Massachusetts)		
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Task Description:	In 1724, Fahrenheit first observed that liquid water could be cooled by a significant amount below its freezing temperature before crystalline ice forms. In 1952, David Turnbull showed that this ability to supercool was a common property of liquids, occurring even in metallic liquids. His experiments demonstrated that a barrier exists to the initial formation of the crystal from the liquid, a process called nucleation. This process is the starting point of not only crystallization, but of most phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. The attainment of an improved ability to control nucleation is of central importance for the development of new materials for advanced applications. In almost all cases the classical theory of nucleation, first developed in the late 19th century, is used to analyze and predict nucleation results. There are, however, serious problems with aspects of that theory in significant regimes of its application. Further, it does not properly describe nucleation when the concentrations of the initial and final phases are different. The purpose of the flight experiment QUASI was to provide a benchmark test of the classical theory and of an extension to that theory to account for different compositions of the initial and final phases. To do this, measurements of the fundamental thermodynamic and kinetic parameters of the supercooled liquids and solid crystallization products are required at the same time that the nucleation rates are measured. Unavoidable convective stirring and the inability to make all of the physical measurements under terrestrial gravity make the microgravity environment of the International Space Station (ISS) essential for such studies. The planned flight experiments would have used the Materials Science Laboratory – Electromagnetic Levitator (MSL-EML) currently being developed by the European Space Agency (ESA). As a bonus, this facility would also have allowed the growth rates of the complex crystal and ordered noncrystallographic phases that form to be measured, allowing theoretical improvements of growth processes, also essential for improved microstructural control. The results of these studies would have lead to an improved ability to develop new materials and phase microstructure and to better tune them to desired applications. As we have already demonstrated in our ground-based studies leading to the flight experiments, new techniques and technologies would also have emerged that would have impacted other areas of science and technology. In 2003, my research group at Washington University, St. Louis, MO, teamed with Dr. Jan Rogers and her team at NASA Marshall Space Flight Center, Huntsville, AL, and Dr. Doug Robinson at the MUCAT beamline of the Advanced Photon Source developed a new technique (Beamline Electrostatic Levitation, or BESL) to obtain rapid structural information at high temperature from electrostatically-levitated samples of Ti-Zr-Ni. We demonstrated that icosahedral short-range ordering in the supercooled liquid catalyzed the formation of a metastable quasicrystal, a noncrystallographic ordered phase that is characterized by long range icosahedral order. In addition to verifying for the first time a half-century old hypothesis about nucleation from a liquid, those studies also demonstrated that this technique could be used to rapidly determine the phase diagrams of high temperature materials. This would be of central interest to the development of new materials for advanced propulsion for aerodynamic and space applications. For flight studies of nucleation, it is essential to first understand all aspects of the nucleation process that are accessible with ground based experiments. As mentioned, the ground-based ESL experiments demonstrated a new connection between structure and the nucleation barrier. In addition, they suggested the occurrence of a metastable critical point in the supercooled liquid, a manifestation of the developing icosahedral order. Evidence suggests a coupling between this higher order liquid ordering transition and the nucleation barrier for the ordered phase. This is a previously unknown type of nucleation, not describable by the classical theory of nucleation and possibly impacting the planned nucleation studies on the ISS. The task of the continuing ground-based research on this contract is the investigation of the icosahedral ordering and associated thermophysical properties in supercooled liquids, primarily of transition metals and their alloys. Measurements are being made of the structure, specific heat, density, viscosity, surface tension and emissivity in the liquids. These data, when combined with the nucleation data, will lead to a better understanding of the complicated nucleation processes that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
	In March, 2005, our post-SCR ISS flight experiment, QUASI, was abruptly cancelled, reflecting a change in the mission of NASA. Since then, we have worked on the analysis and publication of results of ground-based investigations that were carried out in preparation for the expected flight experiments. These include thermophysical property and nucleation data obtained on supercooled liquids using the electrostatic levitation (ESL) facility located at Marshall Space Flight Center and structural data obtained on the MUCAT synchrotron x-ray beamline at the Advanced Photon Source (APS), located at Argonne National Laboratory. The x-ray studies were carried out using the novel facility (BESL, Beamline ESL) conceived and developed by us in 2003 in the ground-based phase of QUASI. The key results from 2006 are listed below. Some aspects of some of the research activities compliment NSF-sponsored research in Kelton's research group at Washington University, sponsored by DMR 03-07410 and DMR 06-06065. The ESL and BESL experiments were made in collaboration with J. Rogers, R. Hyers and A. Goldman. The results obtained from these ground-based studies continue to have a significant scientific impact. 1. Liquid-liquid phase transitions are of scientific interest to a wide community in physics, chemistry, geology, materials science, and more recently, in biology. Theoretical studies indicate that phases with an open tetrahedral molecular coordination are the most likely candidates for such phase transitions and correspondingly experimental and theoretical evidence has been advanced for such a transition in liquid Si. A key prediction is a continuous decrease in the coordination number as the liquid is supercooled. Previous experimental measurements have given contradictory results, with x-ray studies made using conical nozzle levitation methods showing the expected decrease in coordination number with decreasing temperature, while electromagnetic levitation studies showed an increase. Using our BESL technique in-situ high-energy x-ray (125 keV) diffraction studies were made on electrostatic-levitated drops of liquid Si. The data acquired were more accurate than obtained previously and extended to higher momentum transfer. They were obtained with a new area detector that could be operated at a higher rate, allowing a complete diffraction pattern to be taken every 100 ms, allowing data to be obtained at unprecedented levels of supercooling. In contradiction with previous experimental data and with theoretical expectations, these data unequivocally demonstrate that liquid Si maintains a

Task Progress:	constant coordination number independent of supercooling. Over the past year, we have analyzed these experimental data using Reverse Monte Carlo (RMC) methods to obtain atomic configurations and to quantitatively characterize the topological order of the local and medium range structures in the liquids. Many of the features of this structure are the same as those found in structures obtained from ab-initio molecular dynamics simulations. Contrary to widely held expectations based on theoretical studies with the Stillinger-Weber potential, the liquid shows no tendency to develop the tetrahedral diamond cubic structure with decreasing temperature, but instead develops a more ordered A5 structure, indicating that this is the natural structure for the liquid. 2. NASA-sponsored studies of atomic structures obtained from RMC fits to high-angle x-ray scattering data for a Zr59Ti3Cu20Ni8Al10 metallic glass and the associated liquid indicate a growing icosahedral order from the supercooled liquid to the glass. The icosahedral order in the liquid is much less developed than in the glass and is more distorted. These results are in agreement with our NSF-sponsored studies of the time-dependent nucleation rates of the quasicrystal from the glass. A parameter-free analysis of the data revealed an extremely low nucleation barrier to the crystallization to the icosahedral quasicrystal phase, even smaller than the barrier for quasicrystal formation from a supercooled liquid. These results from these structural and nucleation studies provide the first unambiguous experimental evidence that the glass transition is associated with the growth of icosahedral order. Similar results obtained in Zr-Pd liquids and glasses support these conclusions. Further, these studies demonstrate the impact that the local structure of the parent phase can have on phase evolution and microstructural development, in agreement with our earlier work on quasicrystal-forming liquids. 3. A single-cluster analysis of x-ray diffraction data obtained using BESL for Ti, Zr, and Ni supercooled liquids suggested a competition between atomic close packing, favoring icosahedral ordering, and directional bonding in the transition metals. An analysis of the structures obtained by RMC fits to the x-ray data confirmed this conclusion. The growth of icosahedral order was greatest in Ni, the atom closest to having a filled shell. These studies also indicated that much of the icosahedral short range order was highly distorted in all liquids. A decrease in crystalline order was observed, similar to the results for Si. 4. Measurements of the density, viscosity, surface tension, specific heat and viscosity in supercooled Ti-Zr-Ni liquids show anomalous behavior at the same temperature. This is above the recalescence temperature and any conceivable glass transition temperature. Similar properties have been measured in supercooled Zr54Ti8Cu20Al10Ni8 liquids. These data suggest the presence of a novel liquid-liquid phase transition associated with the growth of icosahedral order in the liquid. As discussed in the task description, supercooling experiments suggest a coupling between this phase transition and the nucleation barrier for the nucleation of the primary crystallizing phase. Temperature-dependent emissivity measurements using the ESL are planned in the next few months. These data will allow more quantitative measurements of the temperature dependence of the specific heat, a key parameter for assessing the order of the phase transition. 5. A key feature of the planned ISS experiments was the testing of a new model that we developed to describe the nucleation of phases with a different chemical composition than the parent phase. Our coupled-flux model for nucleation couples the stochastic fluxes of interfacial attachment with those for long-range diffusion. Working with Professor L. Gelb in the Chemistry department at Washington University, Kinetic Monte Carlo (KMC) we made simulations of a lattice gas model (Ising model) to study the effects of this coupling. The simulations confirm several of the distinct features predicted by the coupled-flux model, most importantly, that the chemical composition of regions of the parent phase near sub-critical clusters is closer to that of the new phase than the parent phase. The original coupled-flux model was developed for precipitation from a dilute solution. That assumption is currently being relaxed to develop a more general theory that would be appropriate for the alloys that were planned for the ISS experiments.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
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